

思维链提示

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思维链提示

- 格式: <输入, 思维链, 输出>
- 思维链: 中间推理步骤, 建立输入与输出的联系

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅



思维链提示

➤ 少样本思维链 VS 零样本思维链

➤ 现有主流模型往往可以自发触发思维链模式

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) *The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓*

少样本思维链

(通过**示例**让模型学会生成思维链)

注：通过微调能够让大模型无需特殊提示即可生成思维链

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

零样本思维链

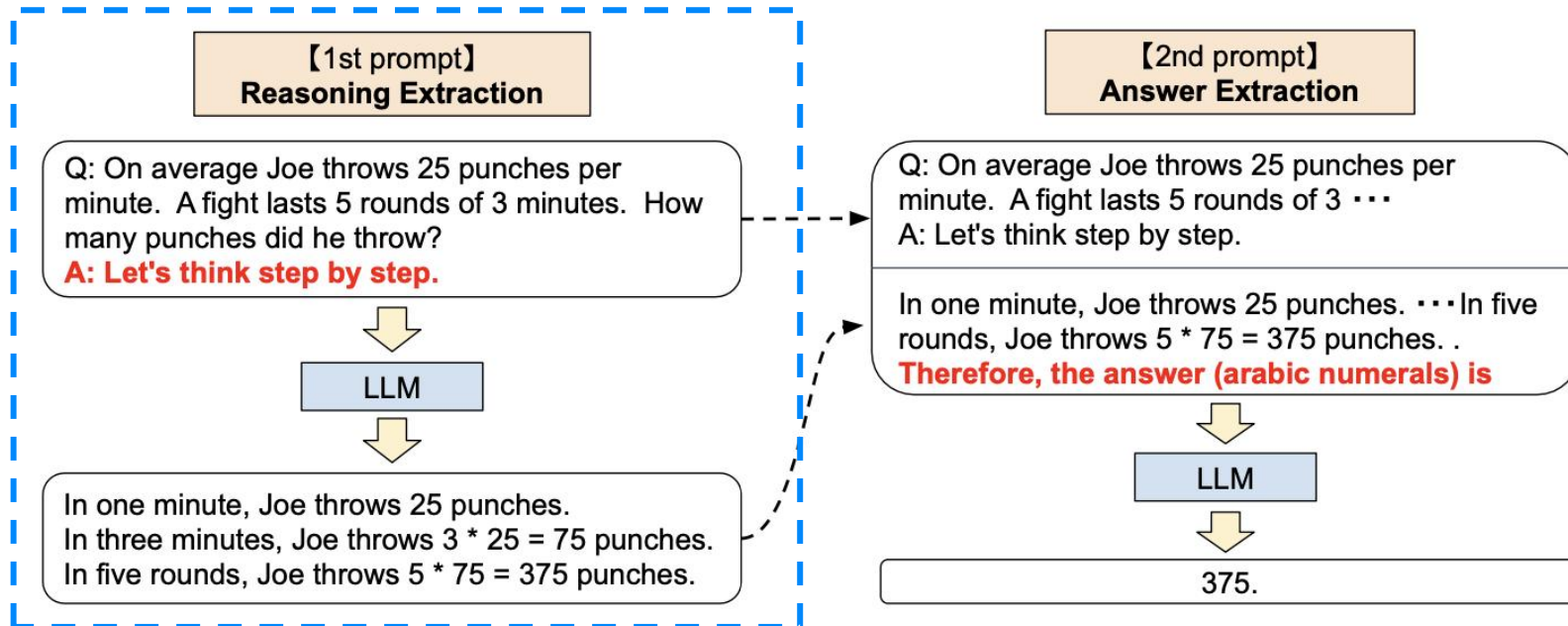
(通过**提示**让模型学会生成思维链)



零样本思维链

➤ workflow

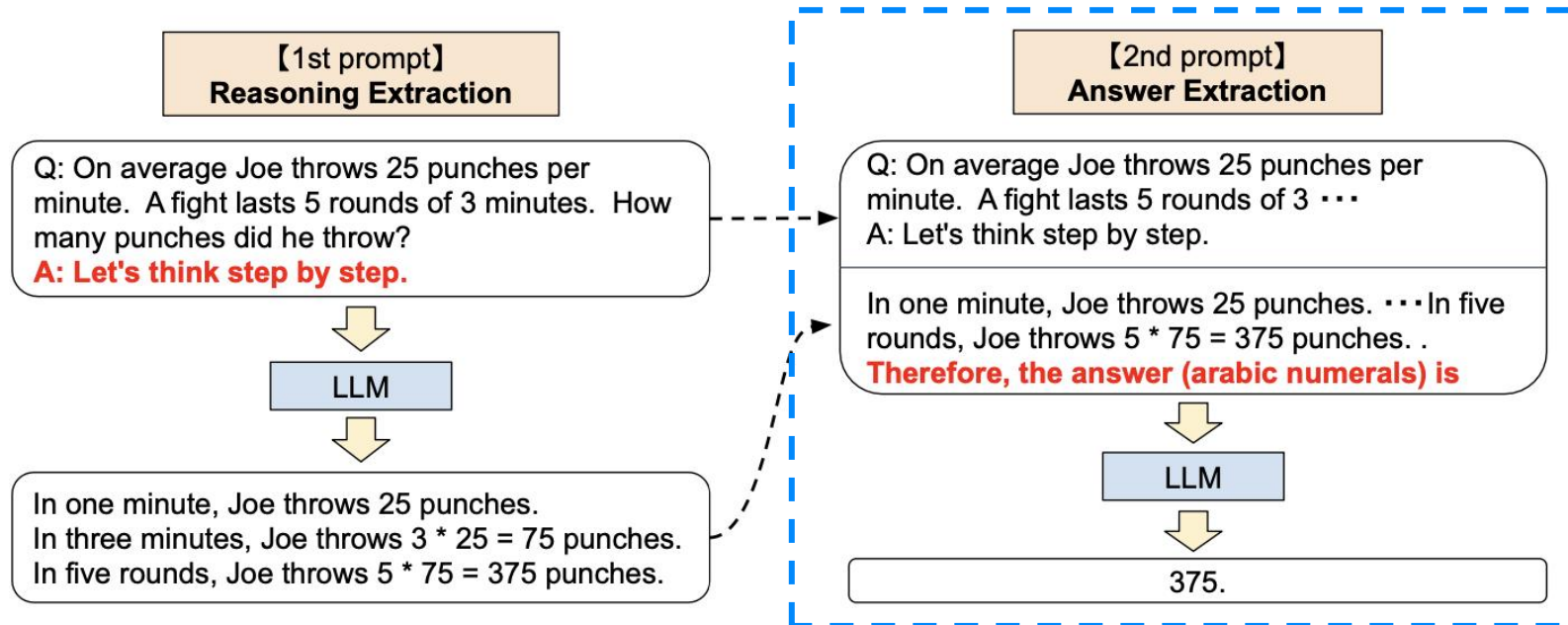
➤ 模型在 Let's think step by step 后生成推理步骤



零样本思维链

➤ workflow

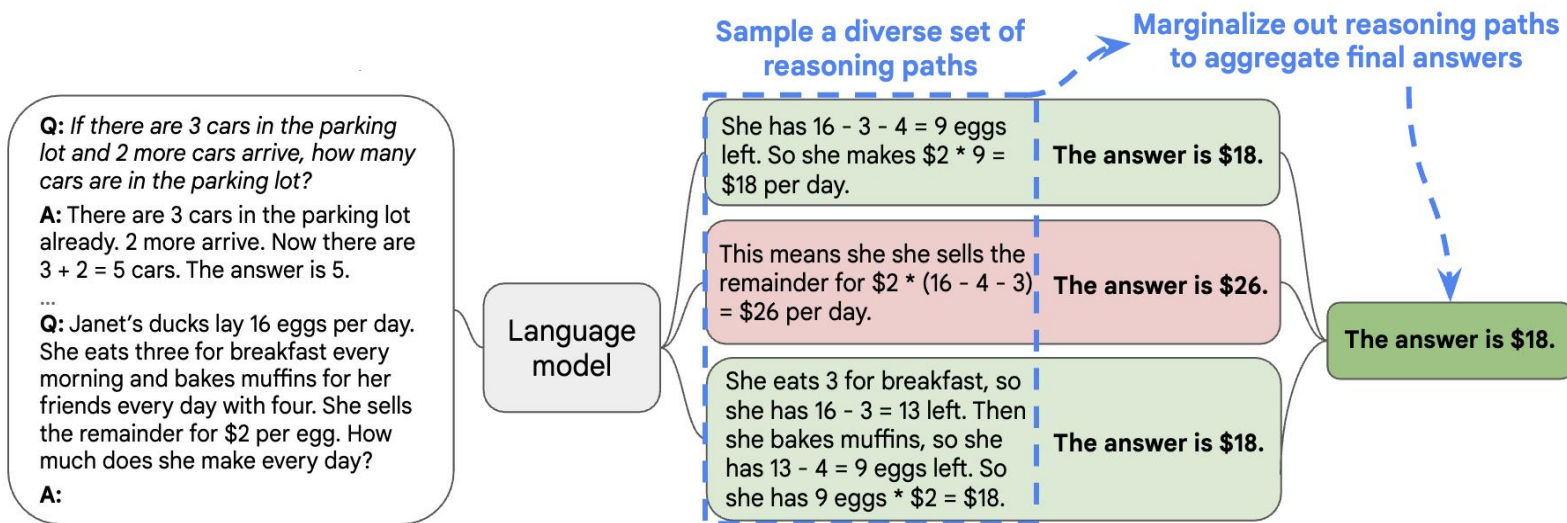
- 模型在 Let's think step by step 后生成推理步骤
- 把推理步骤再作为输入，模型生成答案



改进思维链生成

➤ 基于采样的方法

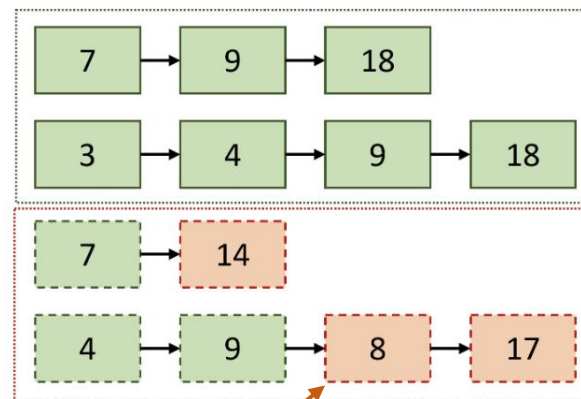
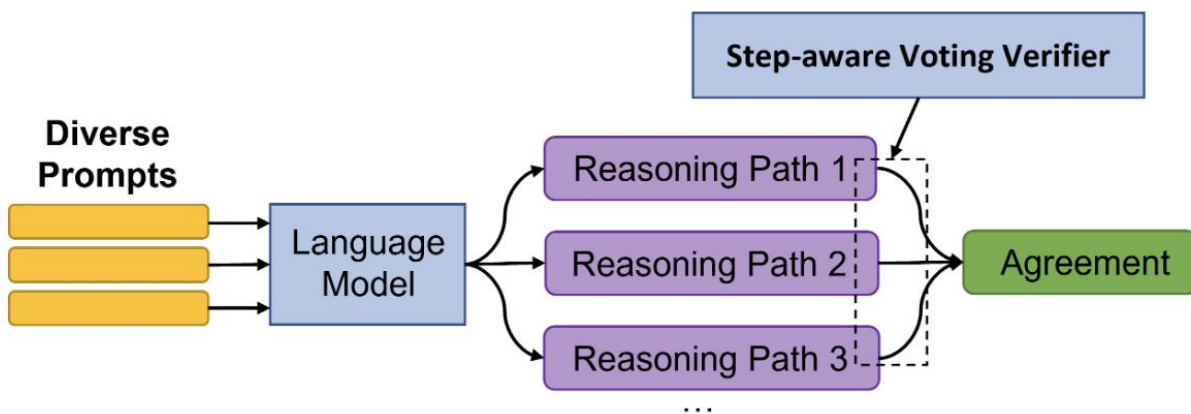
- 问题：在使用单一思维链时，一旦中间步骤出错，容易导致最终答案也出错
- Self-consistency: 生成多条推理路径和对应的答案，然后基于这些答案进行集成（例如选择出现频率最高的答案）并获得最终的答案，也称为majority vote



改进思维链生成

➤ 基于验证的方法

- 问题：思维链的顺序推理本质可能导致推理过程出现错误传递或累计现象
- DIVERSE: 构造针对中间推理步骤的正负数据训练打分模型



推理得到正确
答案的路径

推理得到错误
答案的路径

如果某一步没有在正确路径中出现,
则标记为错误步骤

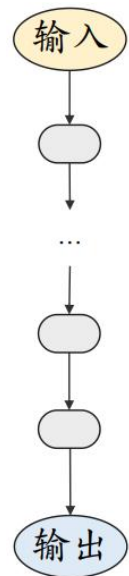


改进思维链结构

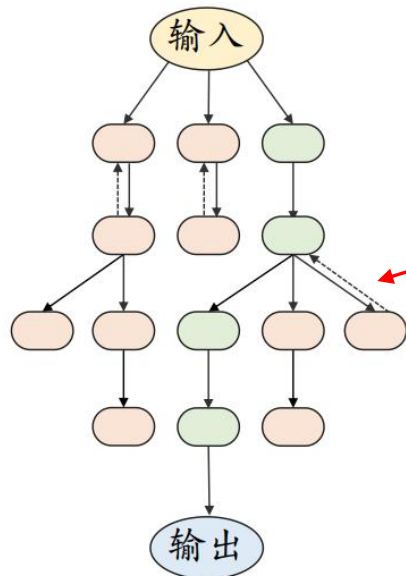
➤ 思维树 (Tree of Thought, ToT)

➤ 每个节点对应一个思考步骤，父节点可以生成多个子节点

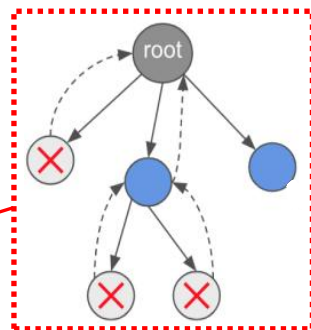
➤ 优点：当子节点出错时具备**回溯**到父节点的能力



思维链



思维树



回溯：从错误子节点返回父节点

→ 推理

--- 回溯

● 正确的想法

● 错误的想法



改进思维链结构

➤ 思维树 (Tree of Thought, ToT)

➤ 算法流程

Algorithm 1 ToT-BFS($x, p_\theta, G, k, V, T, b$)

Require: Input x , LM p_θ , thought generator $G()$ & size limit k , states evaluator $V()$, step limit T , breadth limit b .

$S_0 \leftarrow \{x\}$

for $t = 1, \dots, T$ **do**

$S'_t \leftarrow \{[s, z] \mid s \in S_{t-1}, z_t \in G(p_\theta, s, k)\}$

$V_t \leftarrow V(p_\theta, S'_t)$

$S_t \leftarrow \arg \max_{S \subset S'_t, |S|=b} \sum_{s \in S} V_t(s)$

end for

return $G(p_\theta, \arg \max_{s \in S_T} V_T(s), 1)$

基于广度优先搜索 (BFS) 的思维树

每次保存**b**个
最佳状态

Algorithm 2 ToT-DFS($s, t, p_\theta, G, k, V, T, v_{th}$)

Require: Current state s , step t , LM p_θ , thought generator $G()$ and size limit k , states evaluator $V()$, step limit T , threshold v_{th}

if $t > T$ **then** record output $G(p_\theta, s, 1)$

end if

for $s' \in G(p_\theta, s, k)$ **do** $\triangleright$ sorted candidates

if $V(p_\theta, \{s'\})(s) > v_{thres}$ **then** $\triangleright$ pruning
DFS($s', t + 1$)

end if

end for

基于深度优先搜索 (DFS) 的思维树

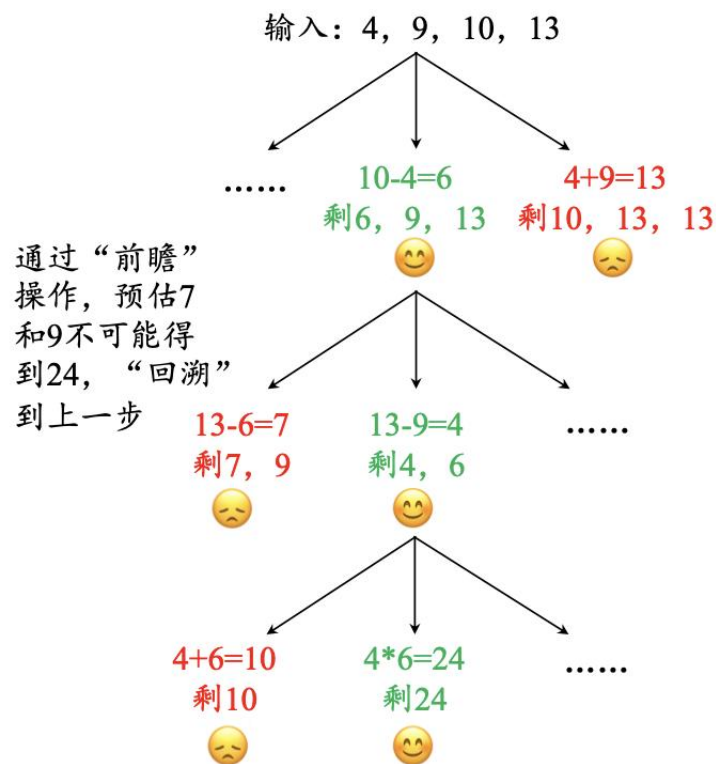
每次对最好的状态进行持续探索



改进思维链结构

➤ 思维树 (Tree of Thought, ToT)

➤ 以 24 点游戏为例 (给定4个数, 通过四则运算得到24)



任务: 给定四个数4、9、10、13, 如何通过加减乘除四则运算得到24?

① 生成多个可能的初始步骤, 例如 $10-4=6$, $4+9=13$

② 每一个步骤都会生成多个下一步骤, 例如 $10-4=6$ 之后可以生成 $13-6=7$ 或 $13-9=4$ 。

③ 对中间步骤进行打分。这里可以对当前思考步骤进行“前瞻”, 例如在当前思考步骤剩下7和9时, 能前瞻性地得知无法得到24, 应该得到一个低分。

④ 如果当前节点不太可能得到最终结果, 那么“回溯”到上一节点, 选择其他路径。例如从 $13-6=7$ 的节点回溯到父节点, 然后前进走到 $13-9=4$ 节点。

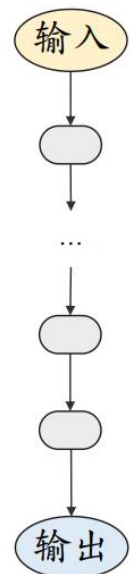


改进思维链结构

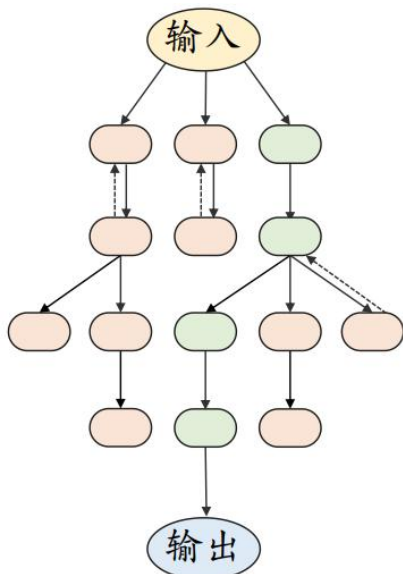
➤ 思维图 (Graph of Thought, GoT)

➤ 任意两个节点可以相连，表示思考步骤之间的依赖关系

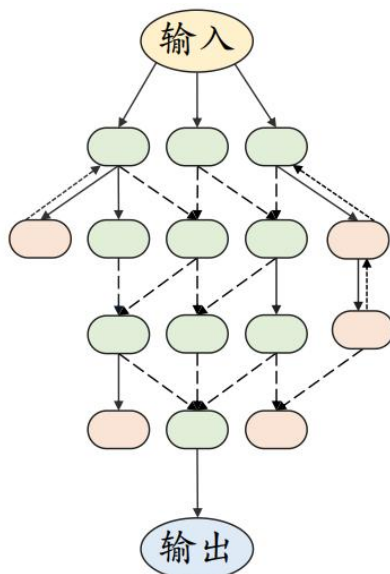
➤ 图结构可以刻画更为复杂的拓扑结构，支持更复杂的推理关系



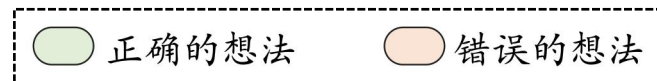
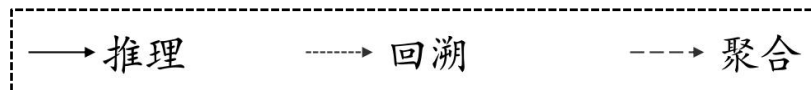
思维链



思维树

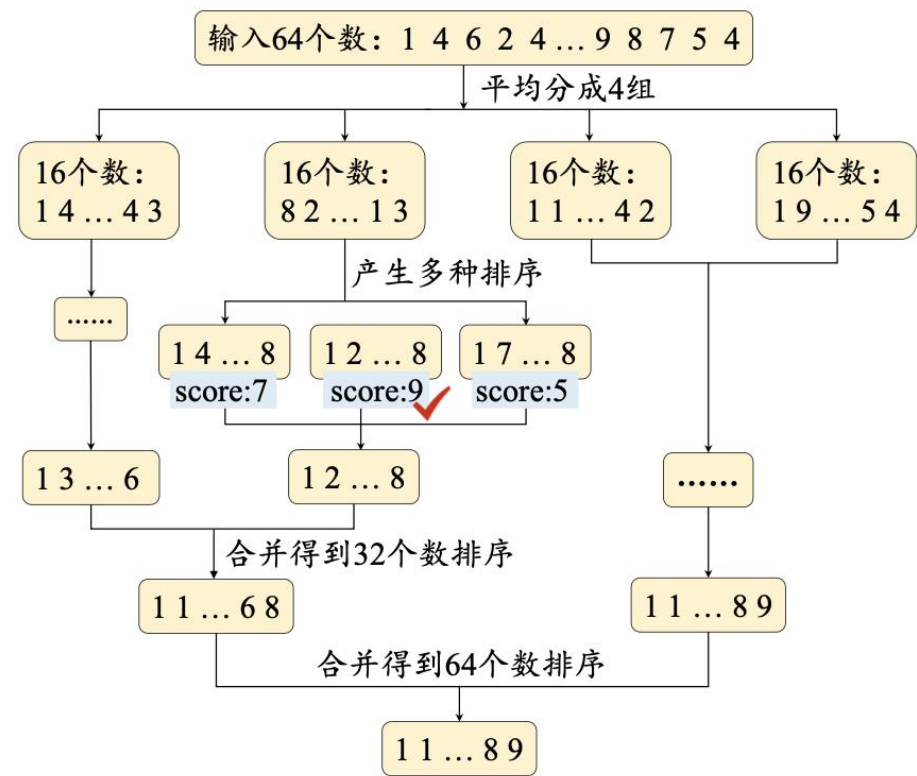


思维图



改进思维链结构

- 思维图 (Graph of Thought, GoT)
- 以含有重复数字的 0~9 数组排序为例



任务: 对64个1~9之间的数字进行排序

- ① 平均分成4组, 每组16个数。
- ② 对每一组而言, 让大语言模型排序多次得到多个可能的结果。
- ③ 对中间步骤进行打分, 保留得分最高的方案。
- ④ 对中间步骤两两合并, 得到最终结果。

注: 在这个场景下, 大语言模型难以对长数组进行准确排序, 但是短数组排序对于模型来说更为简单

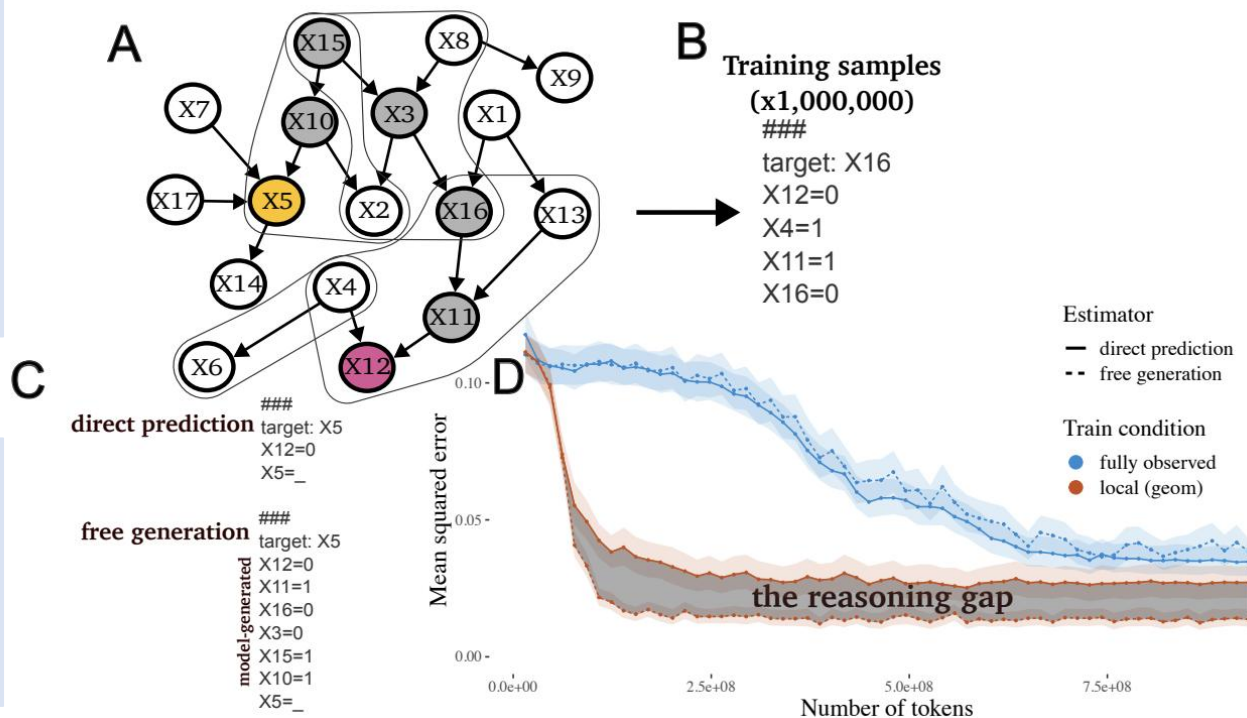


思维链推理能力的来源

- 为什么思维链可以提升复杂推理任务？
- 训练数据中存在相互重叠且互相影响的局部变量空间

A: 贝叶斯网络，红圈表示已观测变量，黄圈表示目标变量，灰圈表示中间变量

C: 比较直接预测目标变量和基于中间变量预测目标变量



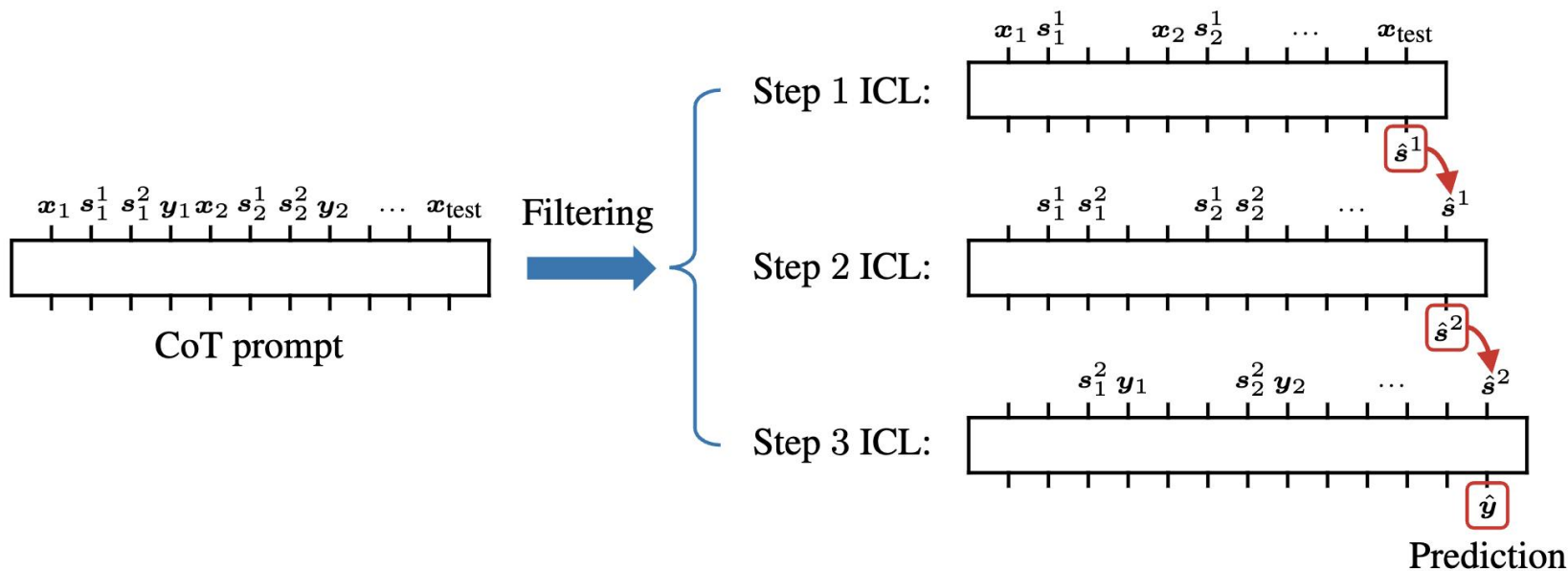
B: 构建具有链式结构的贝叶斯网络，合成包含互相关联的变量的训练样本

D: 当两个变量不经常在数据中共现时，直接预测条件概率与真实概率存在一定偏差；使用中间变量进行推理预测可以减小偏差

思维链推理能力的来源

➤ 思维链可以分解为两个阶段

- 信息聚焦/过滤：模型聚焦于思维链提示中与推理步骤相关的信息
- 上下文学习：基于过滤的信息，通过上下文学习生成一个推理步骤 (即组合函数的单步解)，迭代这一过程获得最终答案 (即组合函数的解)



思维链提示对模型推理的影响

- (少样本) 思维链示例的两个重要组件
 - 符号 (Symbols), 例如数学题的数字、问题中的实体
 - 模式 (Patterns), 例如数学题中的算式、问题的模板

◀ MATHEMATICAL ▶ Solve a grade-school level math reasoning problems
Question: Shawn has five toys. For Christmas, he got two toys each from his mom and dad. How many toys does he have now?
Thought: Shawn started with 5 toys. If he got 2 toys each from his mom and dad, then that is 4 more toys. $5 + 4 = 9$.
Symbols: Numbers: 5, 4, 9
Patterns: Equations: $5 + 4 = 9$. The equations typically appear at the end of the thought, and are almost always involved in generating the final answer.
◀ COMMONSENSE ▶ (SPORTS) Verify the accuracy of a statement linking an athlete with a sport.
Question: Is the following sentence plausible? "Jamal Murray was perfect from the line."
Thought: Jamal Murray is a basketball player. Being perfect from the line is part of basketball.
Symbols: Person and activity: Jamal Murray, Being perfect from the line
Patterns: Consistent sentence structure PERSON belongs to SPORT. ACTIVITY belongs to SPORT , where <i>belongs to</i> is a phrase that connects a sports personality with an activity. The answer is yes if both the person and the activity are associated with the same sport.

不同任务中思维链
示例的符号和模式

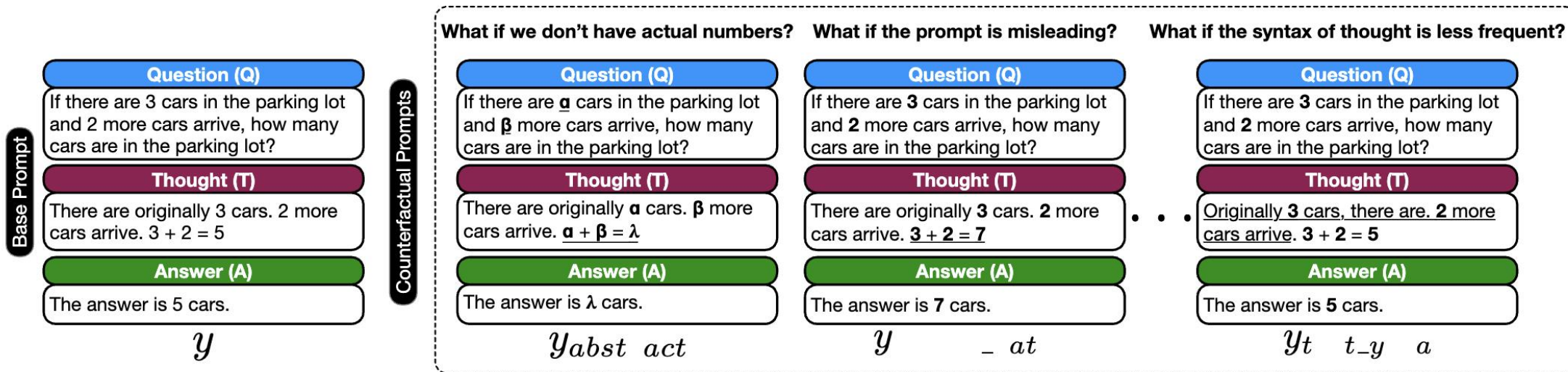


思维链提示对模型推理的影响

➤ 探究思维链对模型推理性能的影响

➤ 反事实提示技术 (类似控制变量法)

➤ 将原始提示的符号或者模式进行修改, 以观察模型性能变化



原始提示

数字替换为希腊字母

使用错误的数字

使用不常见回答句式



思维链提示对模型推理的影响

➤ 探究符号和模式对思维链性能的影响

Question / Thought	Prompt Type	Solve Rate
◀ MATHEMATICAL ▶ (DIRECT = 10.11%, CoT = 27.37%)		
Thought: Shawn started with α toys. If he got β toys each from his mom and dad, then that is λ more toys. $\alpha + \lambda = \pi$.	$C_{\text{symb_abs}}(p)$ (Table 25)	25.70%
Thought: Shawn started with 5.5 toys. If he got 2.5 toys each from his mom and dad, then that is 5 more toys. $5.5 + 5 = 10.5$.	$C_{\text{symb_ood}}(p)$ (Table 30)	28.20%
◀ COMMONSENSE ▶ (SPORTS) (DIRECT = 71.08%, CoT = 93.67%)		
Thought: Jamal Murray is a basketball player. Being ACTIVITY is part of basketball.	$C_{\text{symb_abs}}(p)$ (Table 28)	92.11%
Thought: Adair Foster is a basketball player. Juggling the paper cups is part of basketball.	$C_{\text{symb_ood}}(p)$ (Table 32)	79.72%
◀ COMMONSENSE ▶ (DATE) (DIRECT = 31.61%, CoT = 45.18%)		
Thought: Today is DATE. 24 hours later is one day after today, which would be DATE.	$C_{\text{symb_abs}}(p)$ (Table 24)	37.41%
Thought: Today is 04/30/3069. 24 hours later is one day after today, which would be 04/31/3069.	$C_{\text{symb_ood}}(p)$ (Table 31)	44.50%
◀ SYMBOLIC ▶ (SORTING) (DIRECT = 46.0%, CoT = 60.6%)		
Thought: $\varsigma < \phi < \gamma < \delta < \zeta < \chi < \epsilon < \pi < \upsilon$	$C_{\text{symb_abs}}(p)$ (Table 26)	61.8%
Thought: 11 < 23 < 34 < 48 < 56 < 63 < 72 < 85 < 95	$C_{\text{symb_ood}}(p)$ (Table 33)	80.0%

将符号替换为抽象
的占位符影响很小



思维链提示对模型推理的影响

➤ 探究符号和模式对思维链性能的影响

Question / Thought	Prompt Type	Solve Rate
◀ MATHEMATICAL ▶ (DIRECT = 10.11%, CoT = 27.37%)		
Thought: Shawn started with 5 toys. If he got 2 toys each from his mom and dad, then that is 4 more toys.	$C_{\text{pat_inconsistent}}(p)$ (Table 39)	21.46%
Thought: $5 + 4 = 9$.	$C_{\text{pat_only}}(p)$ (Table 40)	10.01%
Thought: Shawn started with 5 toys. If he got 2 toys each from his mom and dad, then that is 4 more toys. $5 + 4 = 7$.	$C_{\text{pat_wrong}}(p)$ (Table 37)	24.39%
◀ COMMONSENSE ▶ (SPORTS) (DIRECT = 71.08%, CoT = 93.67%)		
Thought: Jamal Murray and being perfect from the line are both part of basketball.	$C_{\text{pat_inconsistent}}(p)$ (Table 45)	79.01%
Thought: Both are part of the same sport .	$C_{\text{pat_only}}(p)$ (Table 41)	74.13%
Thought: Jamal Murray is a soccer player. Being perfect from the line is part of soccer.	$C_{\text{pat_wrong}}(p)$ (Table 46)	46.02%
◀ COMMONSENSE ▶ (DATE) (DIRECT = 31.61%, CoT = 45.18%)		
Thought: Today is 04/19/1969.	$C_{\text{pat_inconsistent}}(p)$ (Table 44)	34.19%
Thought: <calculation> Today = 04/19/1969. 24 hours = 1 day. <output> 04/19/1969 + 1 = 04/20/1969.	$C_{\text{pat_only}}(p)$ (Table 42)	33.52%
Thought: <calculation> Today is 04/19/1969. 24 hours later is one day after today, which <output> would be 03/20/1969.	$C_{\text{pat_wrong}}(p)$ (Table 36)	44.84%
◀ SYMBOLIC ▶ (SORTING) (DIRECT = 46.0%, CoT = 60.6%)		
Thought $9 > 8 > 7 > 6 > 5 > 4 > 3 > 2 > 1$	$C_{\text{pat_inconsistent}}(p)$ (Table 43)	45.0%
Thought: — (similar to DIRECT)	$C_{\text{pat_only}}(p)$	46.0%
Thought: $1 < 2 < 3 < 4 < 7 < 6 < 5 < 8 < 9$	$C_{\text{pat_wrong}}(p)$ (Table 47)	64.8%

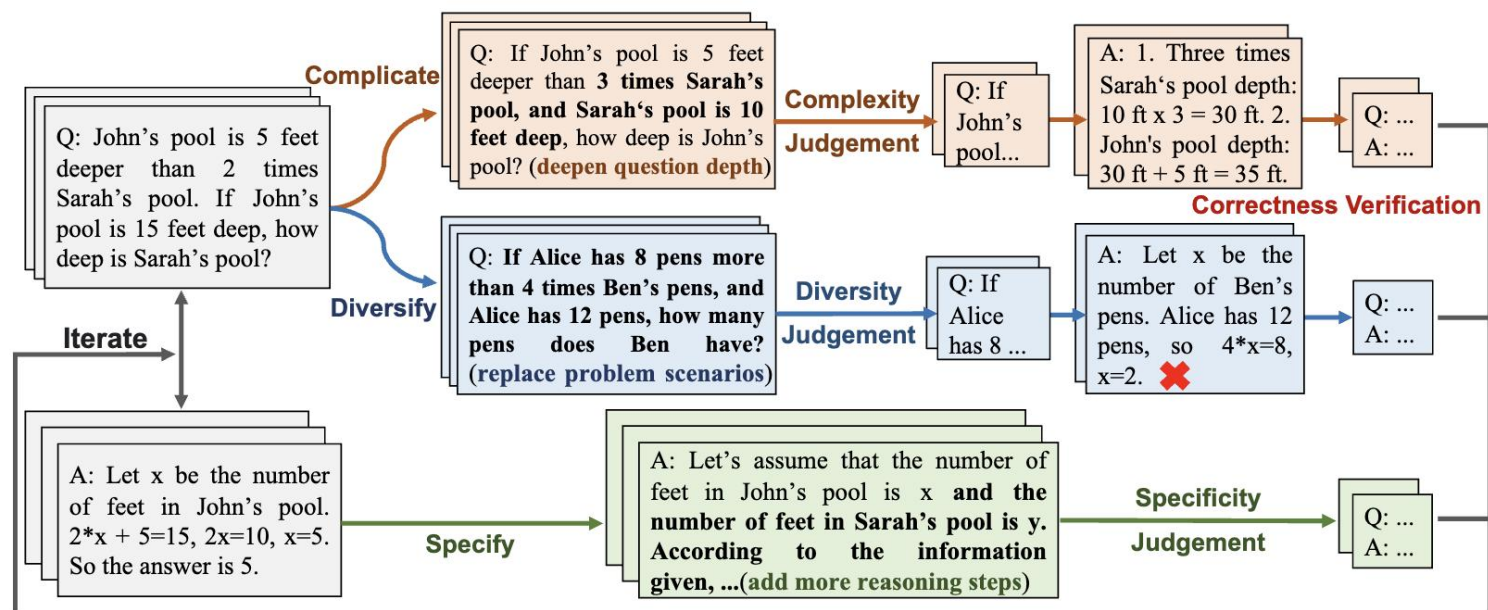
模式与推理过程的一致性和逻辑对思维链性能产生一定影响



如何增强模型的思维链能力?

➤ 思维链数据增强

- **复杂化** (例如增加条件限制、提高问题深度)、**多样化** (例如转换问题背景、提问主题)、**具体化** (例如补充或重写推理步骤)



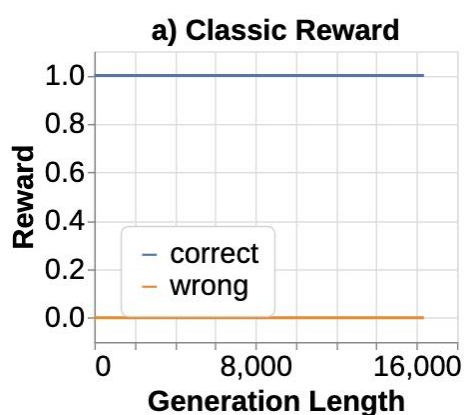
如何增强模型的思维链能力？

➤ 强化学习

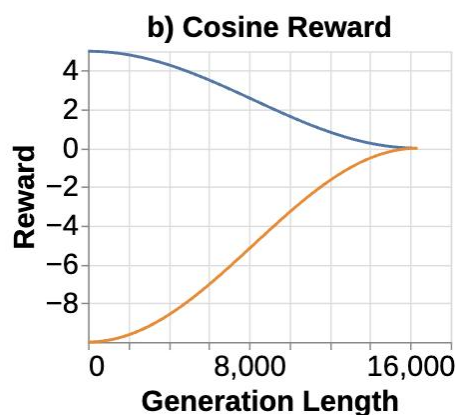
➤ 奖励设计

➤ 传统固定奖励 v.s. Cosine奖励

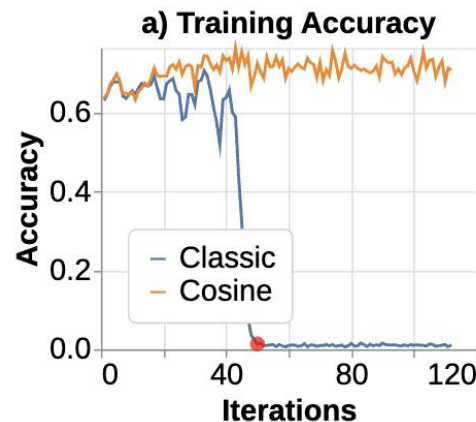
➤ 优点：Cosine奖励使得 RL 更加稳定，帮助控制思维链长度，提升推理准确性



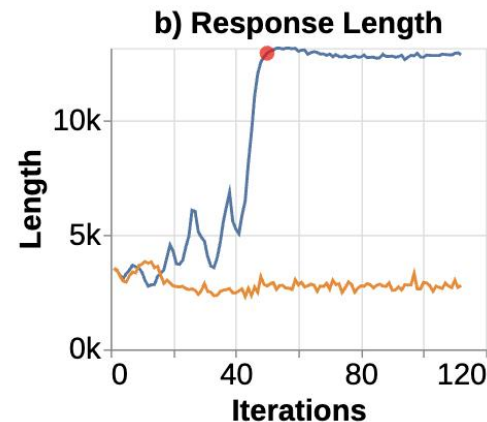
传统固定奖励



Cosine奖励



准确性变化



思维链长度变化



如何增强模型的思维链能力？

- 强化学习
 - 验证器选择
 - 基于规则的验证器 v.s. 基于模型的验证器
 - 优点：基于规则的验证器实现简单，搭配相应的过滤机制效果更好

Prompt Set	Verifier Type	MATH 500	AIME 2024	Theo. QA	MMLU Pro-1k
MATH Baseline		59.4	4.0	25.2	34.6
SFT Initialization		46.6	1.0	23.0	28.3
Unfiltered	Rule-Based	45.4	3.3	25.9	35.1
	Model-Based	47.9	3.5	26.2	40.4
Filtered	Rule-Based	48.6	3.3	28.1	41.4
	Model-Based	47.9	3.8	26.9	41.4



谢谢

